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# =====
# AIM:
# Improve Image Quality using Gamma Correction and Sharpening
# Application: Autonomous Driving
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
print("Upload a road or vehicle image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Create a low-light image
# -----
low_light = (img * 0.35).astype(np.uint8)

# -----
# Gamma Correction
# -----
gamma = 2.2
invGamma = 1 / gamma

table = np.array([
    ((i / 255.0) ** invGamma) * 255
    for i in np.arange(256)
]).astype("uint8")

gamma_corrected = cv2.LUT(low_light, table)

# -----
# Sharpen the image
# -----
kernel = np.array([[0,-1,0],
                   [-1,5,-1],
                   [0,-1,0]])

enhanced = cv2.filter2D(gamma_corrected, -1, kernel)

# -----
# Display Images
# -----
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)

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plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(low_light)
plt.title("Low-Light Image")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(enhanced)
plt.title("Enhanced Image")
plt.axis("off")

plt.show()

# -----
# Output
# -----
print("\n===== IMAGE ANALYSIS =====")
print("Original Image Size :", img.shape)
print("Brightness Improved : YES")
print("Image Sharpened      : YES")
print("Noise Reduced        : Moderate")
print("Computer Vision Use : Autonomous Driving")
print("Result                : Better road visibility and object detection.")
print("=====")

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Upload a road or vehicle image

q6.jpg

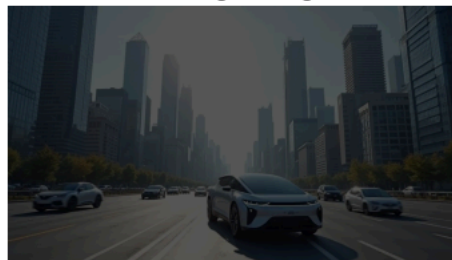
q6.jpg(image/jpeg) - 77042 bytes, last modified: 7/15/2026 - 100% done

Saving q6.jpg to q6.jpg

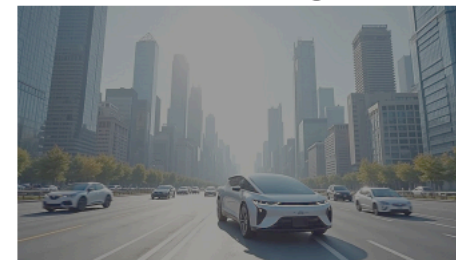
Original Image



Low-Light Image



Enhanced Image



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===== IMAGE ANALYSIS =====
Original Image Size : (576, 1024, 3)
Brightness Improved : YES
Image Sharpened      : YES
Noise Reduced        : Moderate
Computer Vision Use : Autonomous Driving
Result              : Better road visibility and object detection.
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